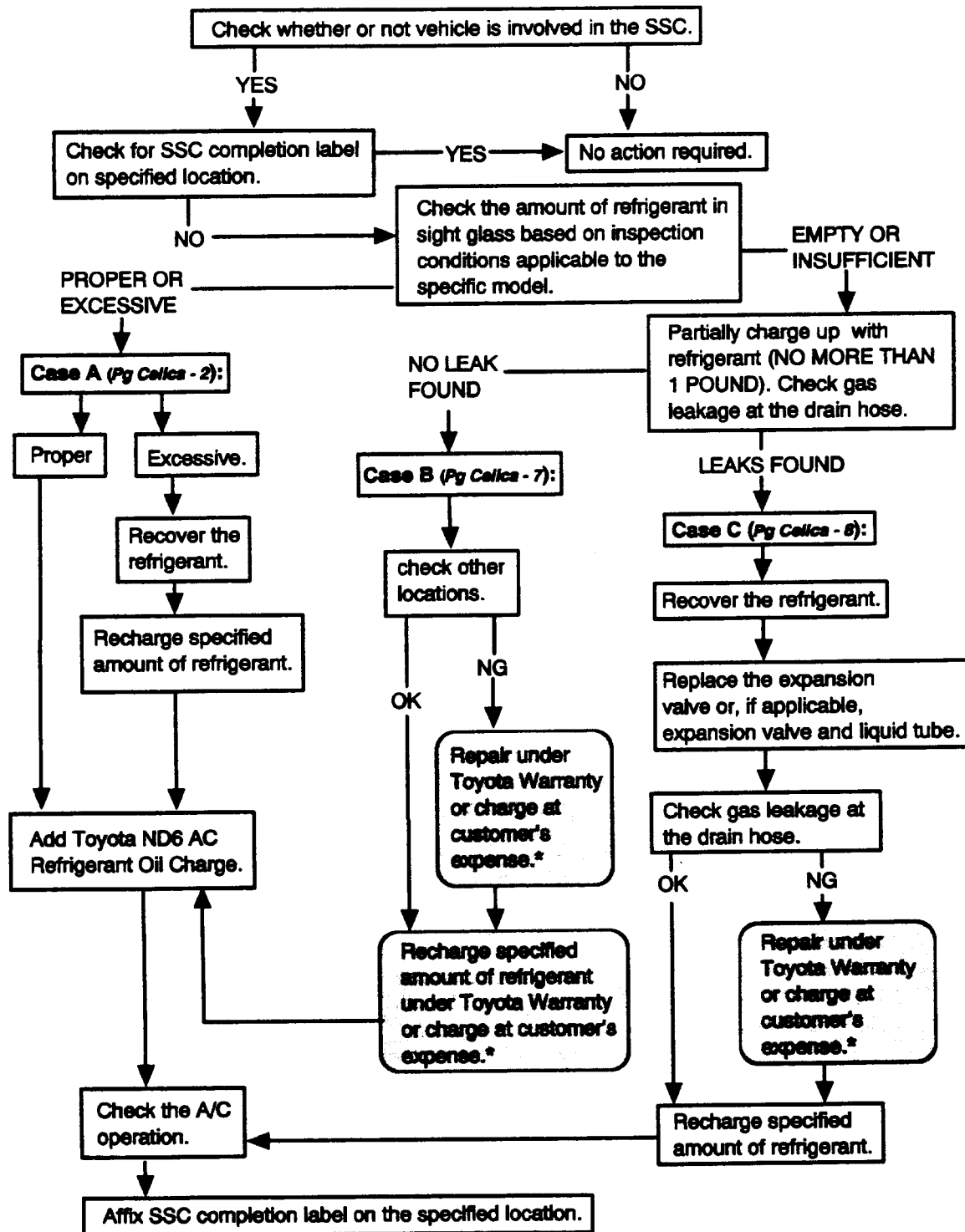


TECHNICAL INSTRUCTIONS FOR CELICA

CELICA

I. OPERATION FLOW CHART FOR SSC P01 - A/C EXPANSION VALVE



* Any repairs and refrigerant charging required for other than the expansion valve are not covered by this Special Service Campaign, and must be covered by basic Toyota Warranty or on a chargeable basis at the customer's expense. Any services requiring customers to pay must be performed with the customer's prior consent.

1. CHECK FOR VIN APPLICABILITY

- (a) Check whether or not the vehicle is affected in the SSC.
- (b) Verify involvement by checking the vehicle identification number on the dashboard plate or certification label on driver's door or door post. (Make sure to check for prior completion label on specified location.)

2. CHECK AMOUNT OF REFRIGERANT FROM SIGHT GLASS

INSPECTION CONDITIONS:

Check the sight glass (**Figure 1**) after maintaining the following conditions for 2 minutes. Perform the check with the cooling fan at a steady speed, not during switching either ON, OFF or between speeds.

CAUTION: Every vehicle shows air bubbles both immediately after starting and when switching from one condition to the other, so it is necessary to wait for stable cooling fan speed to perform the check.

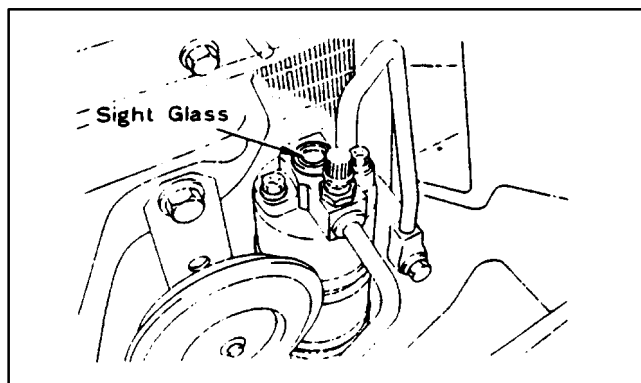


Figure 1

- Conditions:
- (1) Start engine, keep it at 1,500 rpm.
 - (2) A/C switch ON.
 - (3) Blower switch at HI.
 - (4) Air inlet control switch at RECIRC.
 - (5) Temperature switch at Max Cool.
 - (6) Doors fully open.

REFRIGERANT VOLUME INSPECTION STANDARD

Condition in sight glass	Refrigerant Quantity	Symptom	Service Procedure
	Proper	No air bubbles seen in refrigerant flow. When engine speed is raised gradually from idle to 1,500 rpm, the bubbles disappear, making the refrigerant transparent.	Case A
 (Clear) Compressor operating	Excessive	No air bubbles seen in the refrigerant flow.	
 (Clear) Compressor not operating	Empty	No air bubbles seen in the refrigerant flow.	Case B/C
	Insufficient	Air bubbles seen continuously in the refrigerant flow.	

3. WORK PROCEDURE FOR CASE A:

Depending on the refrigerant volume, proceed as follows:

REFRIGERANT VOLUME CORRECT:

Proceed to **ADDITION OF TOYOTA ND6 AC REFRIGERANT OIL CHARGE.**

REFRIGERANT VOLUME EXCESSIVE:

Recover refrigerant, recharge with 720 +/- 50 g (25.4 +/- 1.8 oz,
1.6 +/- 0.1 lb)

Proceed to **ADDITION OF TOYOTA ND6 AC REFRIGERANT OIL CHARGE.**

ADDITION OF TOYOTA ND6 AC REFRIGERANT OIL CHARGE

NOTICE: Before starting the operation, please read and observe all cautions and warnings written on the refrigerant oil charge can.

Equipment needed for this operation:

- (i) 1 can of Toyota ND6 AC Refrigerant Oil Charge. (Part No. 00263-09106)
- (ii) AC R-12 vacuum pump.
- (iii) AC Oil Injection Kit (Supplied to each dealer). Special Service Tool Part No. TOY219721.
- (iv) Shop towel.

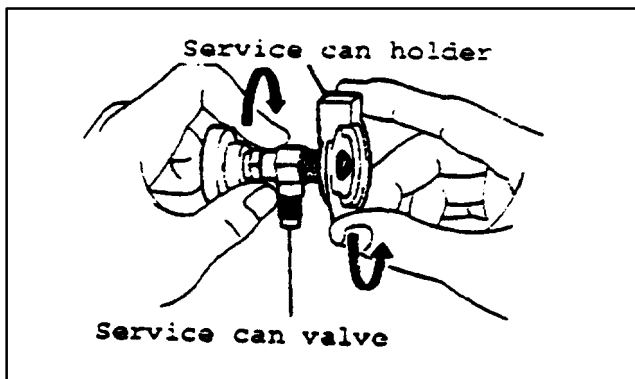


Figure 2

- (1) Before servicing, verify the operation of the Special Service Tool by turning the service can valve counterclockwise and remove it from the service can holder.
(Figure 2)

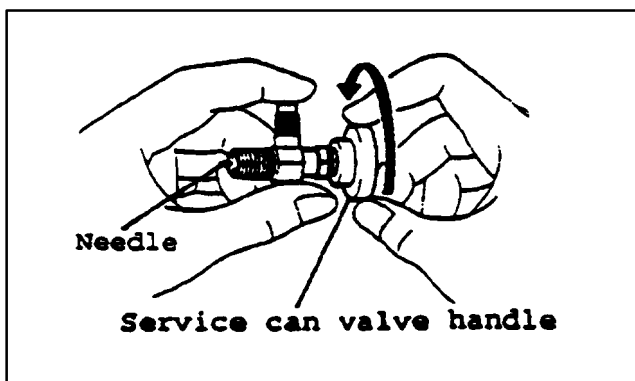


Figure 3

- (2) Turn the service can valve handle clockwise until you can't turn it any more. Verify that the valve puncture needle protrudes fully.
(Figure 3)

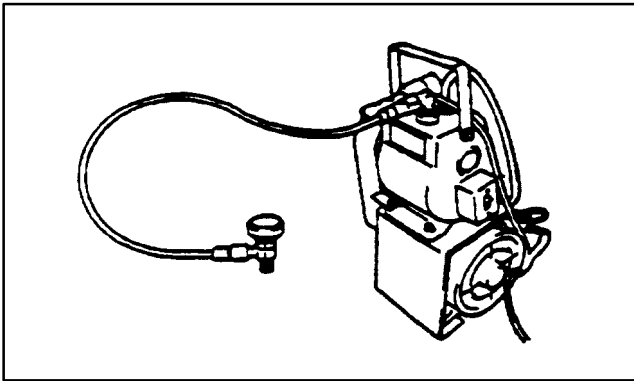


Figure 4

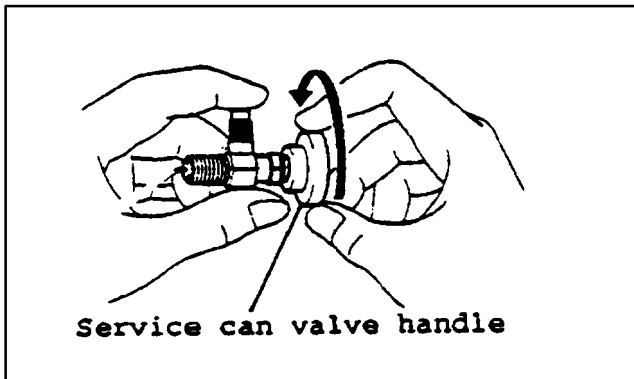


Figure 5

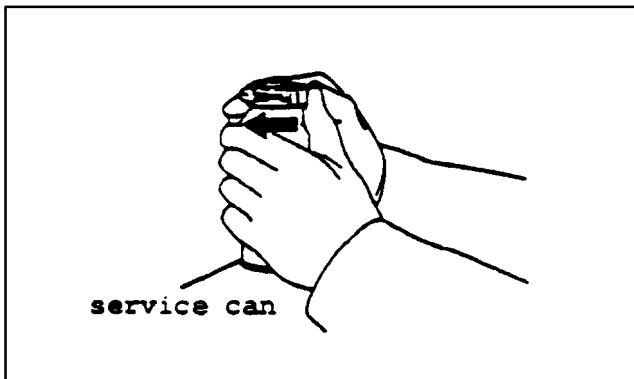


Figure 6

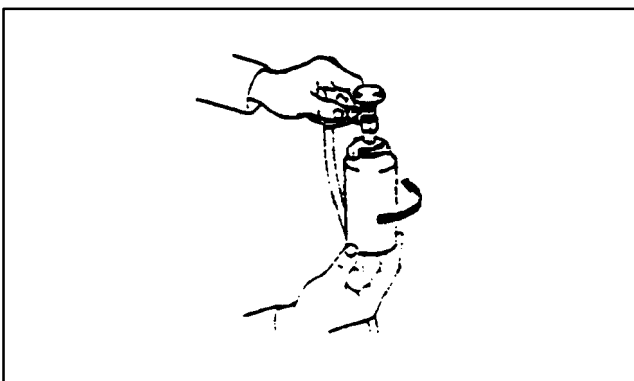


Figure 7

- (3) When performing the AC refrigerant oil injection for the first time, or if the oil injection service can valve and hose assembly from the Special Service Tool has been accidentally opened to the atmosphere, it should be evacuated first.

Evacuation Procedure:

- (i) To evacuate the service can valve and hose assembly, first attach the smaller hose connection to the service can valve.
- (ii) Next, attach the larger hose connection with self-sealing stop valve end to the AC vacuum pump.

(Figure 4)

NOTE: Make sure both connections are tightly secured.

- (iii) Turn the service can valve handle clockwise until you can't turn it any more. **(Figure 5)**
- (iv) Start the vacuum pump and allow it to run for approximately 2 minutes, or until the vacuum gauge on the vacuum pump shows no more movement. Then quickly remove the hose connection from the vacuum pump **BEFORE** switching off the pump to keep air from entering the hose.

NOTE: If vacuum pump is turned off first before the hose is disconnected, the hose will be contaminated with air and the evacuation procedure will have to be repeated.

For Regular/Subsequent Injection Service:

- (1) After evacuation, with the service can valve still seated, install the service can holder on the refrigerant oil charge can. **(Figure 6)**
- (2) Turn the oil charge can with holder clockwise to the service can valve until it cannot turn any further. **(Figure 7)**

NOTE: Be sure to turn the service can valve quickly onto the service can holder until it seats snugly against the refrigerant oil charge can. This is to prevent refrigerant from escaping to the atmosphere when the valve needle punctures the top of the can.

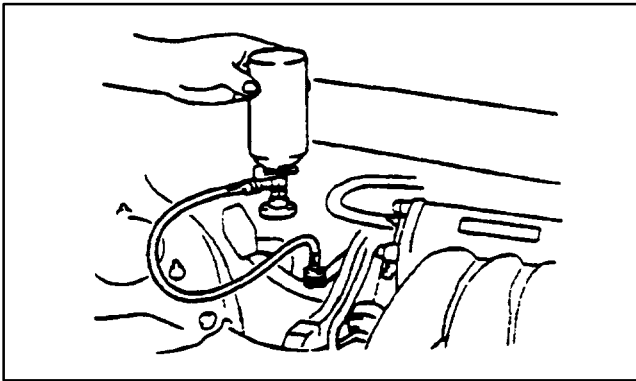


Figure 8

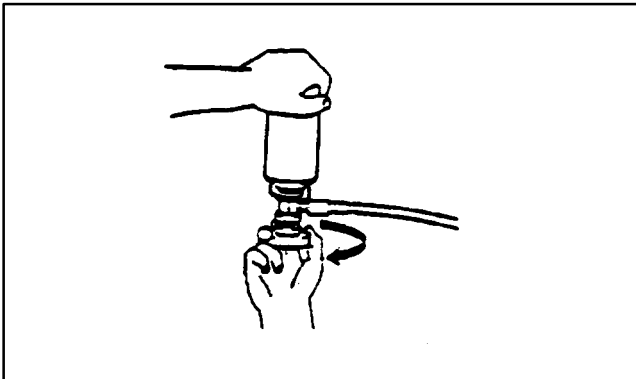


Figure 9

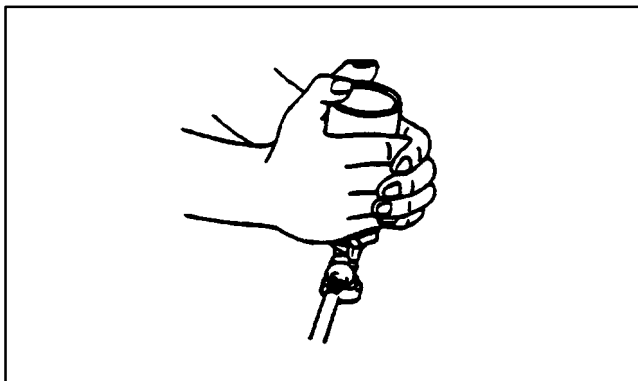


Figure 10

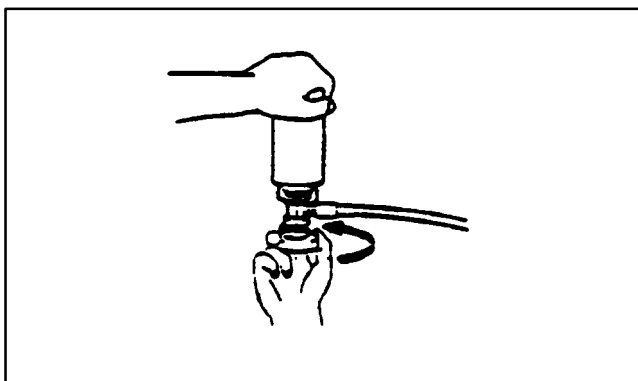


Figure 11

- (3) Remove the cap from the low pressure charge valve on the vehicle. Attach the larger hose connection to the low pressure charging valve.

NOTE: Use a shop towel when doing this to shield your hands from any refrigerant that may escape the vehicle's A/C system.

- (4) Run the engine at approximately 1,500 rpm, and operate the air conditioning system (A/C switch ON, temperature control switch set at Max Cool and the blower fan on HI).
- (5) Turn the refrigerant oil charge can upside down to make sure that oil is fully charged by gas pressure. **(Figure 8)**

- (6) Turn the service can valve handle counterclockwise to open the refrigerant oil charge can. **(Figure 9)**

- (7) Warm the refrigerant oil charge can with your hands or a warm towel so that the refrigerant oil charge can is emptied more easily. **(Figure 10)**

- (8) When charging is complete, turn the service can valve handle clockwise to shut it off. **(Figure 11)**

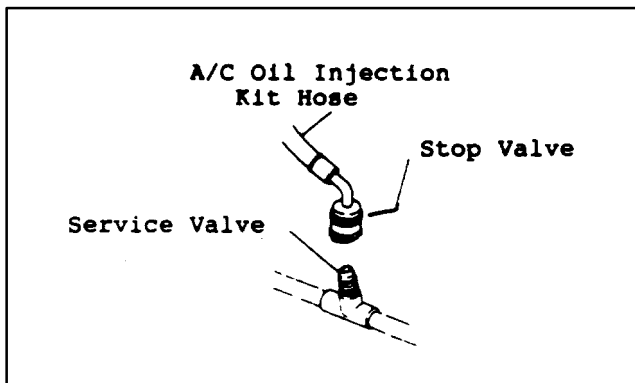


Figure 12

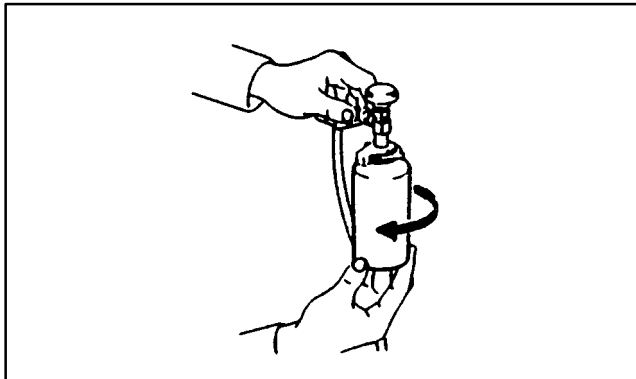


Figure 13

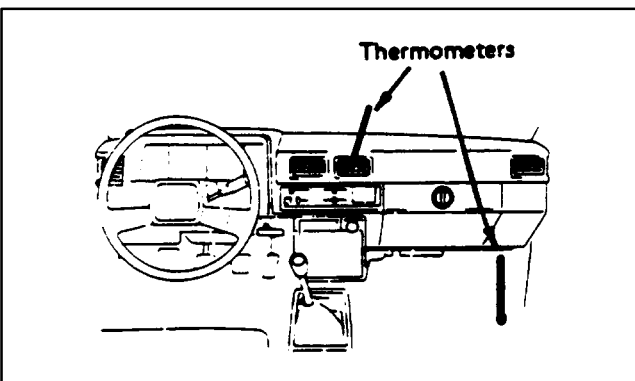


Figure 14

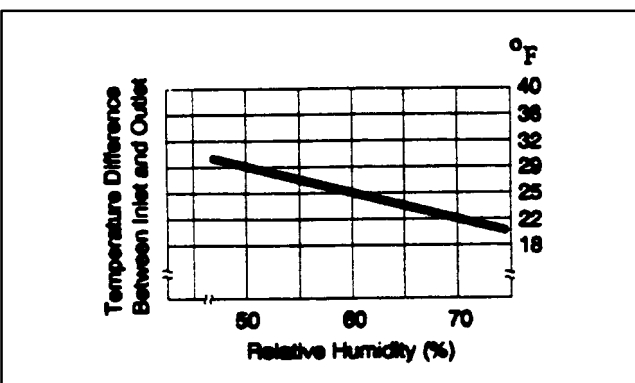


Figure 15

- (9) Stop the engine and disconnect the charging hose from the vehicle's low pressure charging valve. **(Figure 12)**
- (10) Turn the service can valve counter-clockwise and remove it from the service can holder and the refrigerant oil charge can. **(Figure 13)**
- (11) Remove the service can holder from the refrigerant oil charge can by moving the lever counterclockwise.

Caution: Do not open service can valve between services. If the valve is opened, the service can valve and hose assembly will have to be evacuated again.

- (12) Operate the air conditioner.
Set the following conditions:
 - * Start engine, keep it at 1,500 rpm.
 - * A/C switch ON.
 - * Blower switch at HI.
 - * Air inlet control switch at RECIRC.
 - * Temperature switch at Max Cool.
 - * Doors fully open.
- (13) Position thermometers. **(Figure 14)**
 - (i) Place a thermometer in the cool air outlet.
 - (ii) Place a second thermometer near the inlet of the cooling unit.
- (14) Check performance of A/C system.
 - (i) Measure the temperatures at the air outlet and air inlet, and calculate the difference.
 - (ii) Check that the intersection of the relative humidity and temperature difference is above the line shown. If the intersection is above the line, cooling performance is satisfactory. **(Figure 15)**
- (15) If A/C performance is unsatisfactory, check other components and repair according to repair manual under warranty or at customer's expense (with consent).
- (16) Install SSC completion label as directed on Page Celica-11.

4. GAS LEAK INSPECTION FROM EXPANSION VALVE OF A/C COOLING UNIT.

There are 2 further conditions: **CASE B** and **CASE C**.

To determine which work procedure to use (**CASE B** or **CASE C**), please use the selection procedure below:

PARTIAL CHARGING OF REFRIGERANT

If there is no refrigerant in the A/C system, partially charge it with refrigerant in order to perform the following inspection.

Partial Charge Amount: **No more** than 450 g (1 lb)

GAS LEAK INSPECTION AT DRAIN HOSE

Use a leak detector to determine whether gas is leaking out of the A/C drain hose from the expansion valve. (**Figure 16**)

JUDGEMENT

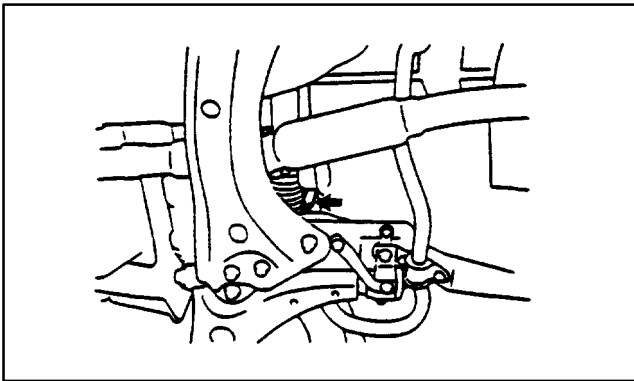


Figure 16

CASE B: Gas is not detected leaking out of the drain hose end. Follow procedures under **CASE B**. (Page Celica-7)

CASE C: Gas is detected leaking out of the drain hose end. Follow procedures under **CASE C**. (Page Celica-8)

5. WORK PROCEDURE FOR CASE B.

GAS LEAK INSPECTION FOR OTHER LOCATION

Check whether A/C is leaking from anywhere other than the drain hose end. Use a leak detector to determine whether gas is leaking from the locations shown by the arrows below.

Inspection locations for vehicle.

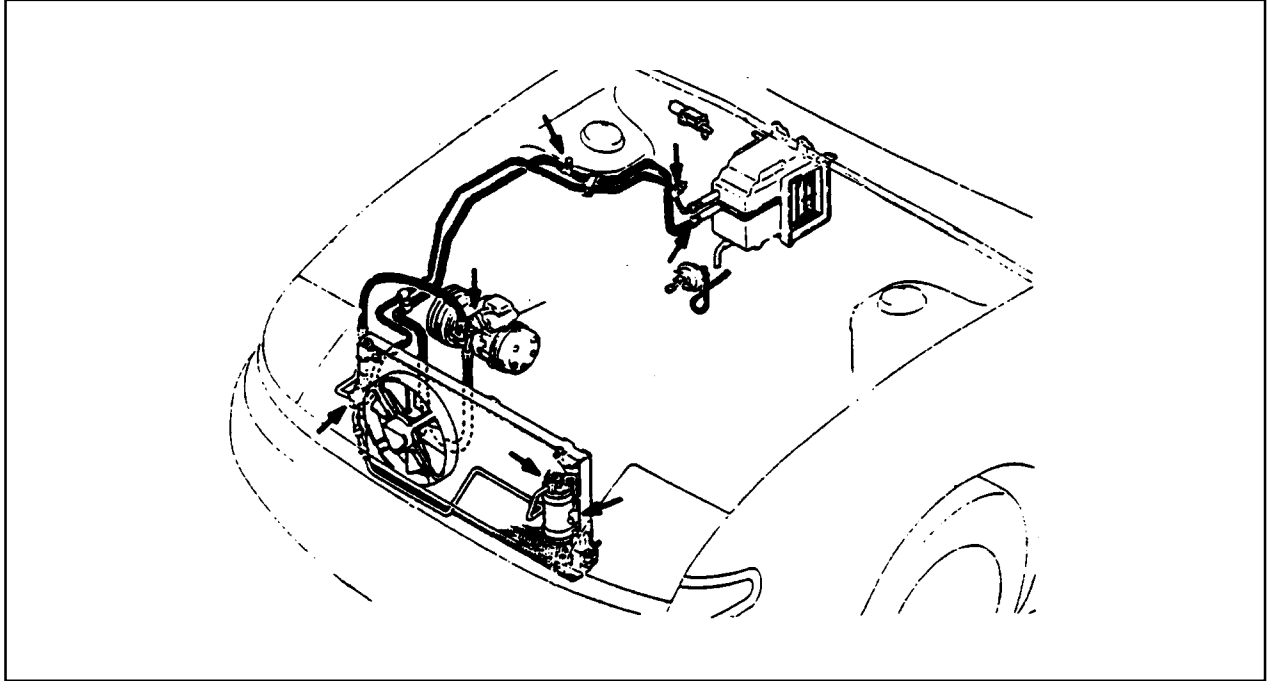


Figure 17

INSPECTION RESULT AND ACTION

OK (No leaks found): Confirm full refrigerant charge. Cost of recharging will be covered by Toyota Warranty or at customer's expense (with consent) as applicable.

Add Toyota ND6 AC Refrigerant Oil Charge.

See **WORK PROCEDURE FOR CASE A** on Page Celica-2.

NG (Leaks found): These leaks are either:

- (i) Covered by Toyota Basic Warranty Policy, or
- (ii) Charge at customer's expense. (Ensure that customer gives prior consent before proceeding with the repair.)
- (iii) Following completion of repair, add the Toyota ND6 AC Refrigerant Oil Charge as shown on Page Celica-2.

6. WORK PROCEDURE FOR CASE C

NOTICE: When performing this procedure:

- * Take care not to scratch or break connectors, or cut any wire harnesses.
- * Press connectors together until you hear a solid clicking sound.

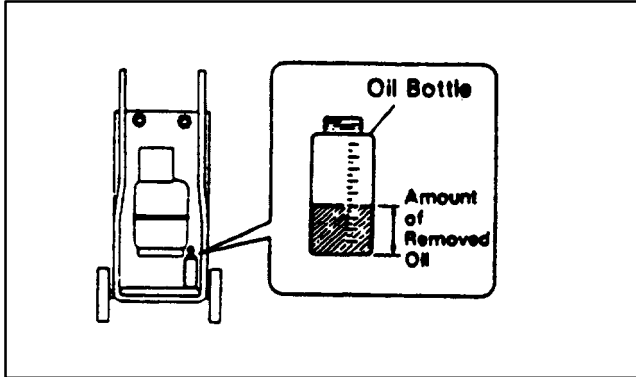


Figure 18

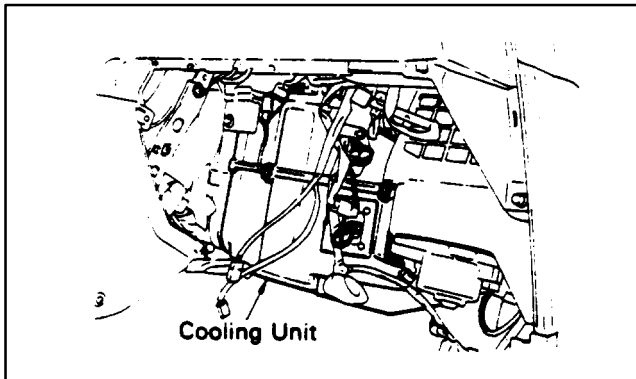


Figure 19

RECOVERY OF REFRIGERANT FROM A/C SYSTEM

When replacing parts, recover the refrigerant. Always use a recovery machine when recovering the refrigerant. (See Preparation and Precaution.)

NOTICE: When the refrigerant is recovered, the compressor oil is also being recovered with the refrigerant. When recharging a system that has been evacuated, **DO NOT** reuse the old compressor oil. Measure the amount of compressor oil removed and install the same amount of **NEW Toyota ND OIL 6 Compressor Oil** into the system. (Figure 18)

EXPANSION VALVE REPLACEMENT

- (1) Disconnect negative cable from battery.
- (2) Disconnect suction tube from cooling unit outlet fitting.
- (3) Disconnect liquid tube from cooling unit inlet fitting.

NOTE: Cap the open fittings immediately to keep moisture out of the system.

(Figure 19)

- (4) Remove grommets from inlet and outlet fittings.
- (5) Remove glove box and reinforcement.
- (6) Disconnect connectors.
- (7) Remove cooling unit by removing three nuts and four bolts.

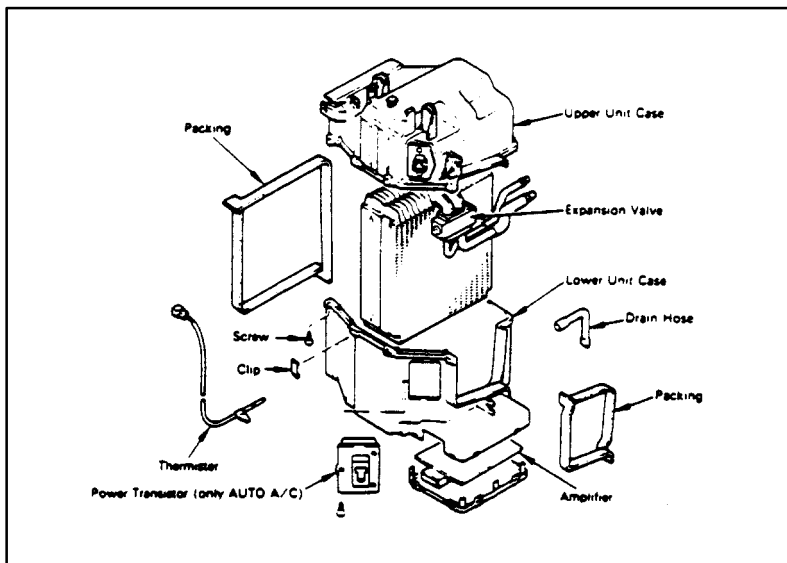


Figure 20

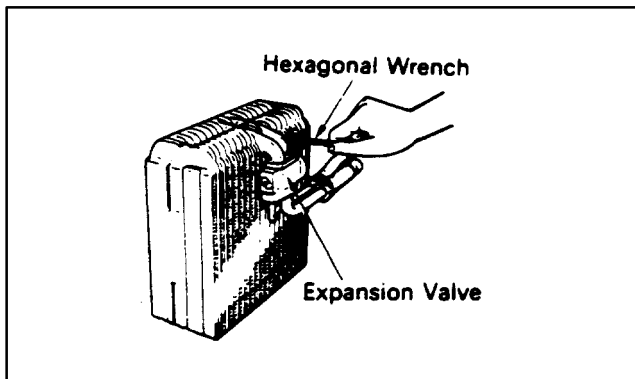


Figure 21

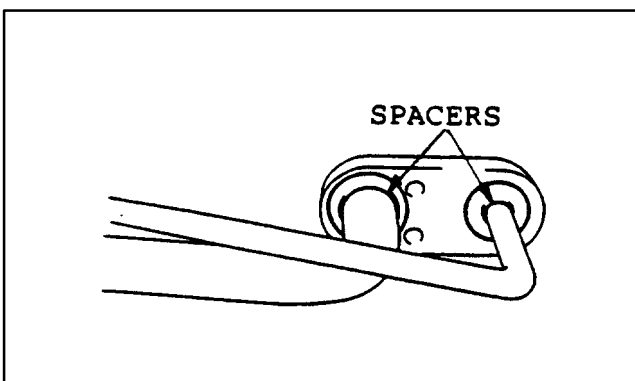


Figure 22

(8) Remove lower and upper unit cases. **(Figure 20)**

- (i) Disconnect connectors.
- (ii) Remove four clips.
- (iii) Remove four screws.
- (iv) Remove upper unit case.
- (v) Remove thermistor with thermistor holder.
- (vi) Remove lower unit case.

(9) Remove expansion valve, and liquid and suction tube.

- (i) Remove the packing and heat sensing tube from suction and liquid tubes.
- (ii) Remove the expansion valve from the evaporator by removing the two bolts using a hexagon wrench. **(Figure 21)**
- (iii) Replace the expansion valve, or if required, the expansion valve and liquid and suction tube based on applicability shown in Appendix.

Replace the tube fitted with zinc plated steel spacers (gold color) at the tube end. (Figure 22)

NOTE: Be sure to replace the O-rings with new ones as listed below.

- 2 O-rings between expansion valve and evaporator.
- 2 O-rings between expansion valve and liquid and suction tube.
- 2 O-rings where liquid and suction tube joins tubes on vehicle side.

Apply a few drops of compressor oil to O-ring fittings for easy tightening and to prevent leaking of refrigerant gas.

- (10) Reinstallation of cooling unit.
- (i) Connect the expansion valve, suction and liquid tubes to the evaporator.
Torque the bolt.
- NOTE:** Be sure that the O-rings are positioned on the tube fitting.
- (ii) Install lower unit case to the evaporator.
 - (iii) Install thermistor to the evaporator.
 - (iv) Install upper unit case.
 - (v) Install four screws.
 - (vi) Install four clips.
 - (vii) Connect connectors.
 - (viii) Reinstall cooling unit with three nuts and four bolts.
- (11) Reinstall connectors.
- (12) Reinstall glove box and reinforcement.
- (13) Reinstall grommets on inlet and outlet fittings.
- (14) Reconnect liquid tube to cooling unit inlet fitting.
- (15) Reconnect suction tube to cooling unit outlet fitting.
- (16) Reconnect negative cable to battery.
- (17) Evacuate air from air conditioning system.

REFRIGERANT LEAK INSPECTION

Use a leak detector to check for refrigerant leaks as instructed below.

Evacuation Process

1.
Evacuation

10 minutes
or more

2.
Airtight
Check

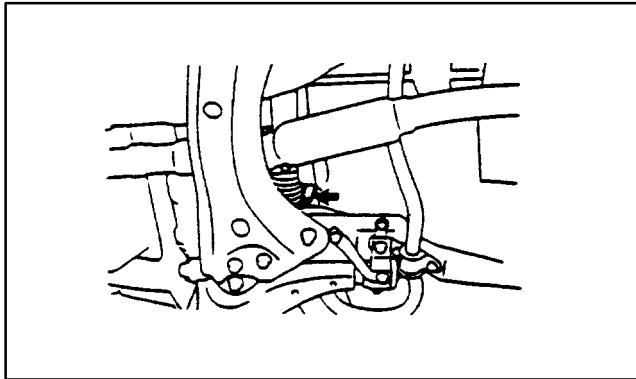
Leave for
5 minutes

3. Partial
Charging of
Refrigerant

Add no more
than 1 lb

4. Refrigerant
Leak Check at
Drain Hose

Use a leak
detector to
check



Refrigerant leakage from the evaporator or from the expansion valve location can be inspected at the drain hose. **(Figure 23)**

Figure 23

INSPECTION RESULT AND ACTION

- | | | |
|-----|----------------------|---|
| (1) | OK (No leaks found): | Charge the refrigerant. |
| (2) | NG (Leaks found): | These leaks are either: <ul style="list-style-type: none"> * Covered by Toyota Basic Warranty Policy, or * Charge at customer's expense (ensure that customer gives prior consent before repair.) |

REFRIGERANT CHARGING VOLUME

Specified amount: 720 +/- 50g (25.4 +/- 1.8 oz, 1.6 +/- 0.1 lb)

A/C OPERATION INSPECTION

Follow procedures shown on Page Celica-5 (12).

TORQUE SPECIFICATIONS

- | | |
|---------------------------------|-------------------------------|
| * Expansion Valve x Evaporator: | 5.4 N-m (55 Kgf-cm, 48 in-lb) |
| * Liquid Tube x Cooling Unit: | 13 N-m (135 Kgf-cm, 10 ft-lb) |
| * Suction Tube x Cooling Unit: | 32 N-m (325 Kgf-cm, 24 ft-lb) |

SSC COMPLETION LABEL INSTALLATION

- (i) After completing repair and before returning vehicle to the owner, a SSC completion label that is enclosed in the owner's letter must be affixed to the left front door hinge post near the check strap.
- (ii) The label is to be filled out as follows:

Write in SSC P01

SSC	Date
TOYOTA MOTOR SALES, U.S.A., INC.	
DEALER CODE NO.	
00410-01917	

Write in date of Repair

Write in your Dealer Code

- (iii) Additional SSC completion labels, in sheets of 50 (P/N 00410-01917), may be ordered through the non-parts system on a 1450 order form or through the TDN system.